

TPS63900 1.8-V to 5.5-V, 75-nA I_Q Buck-boost Converter with Input Current Limit and DVS

1 Features

- Input voltage range: 1.8 V to 5.5 V
- Output voltage range: 1.8 V to 5 V (100-mV steps)
 - Programmable with external resistors
 - SEL pin to toggle between two output voltage presets
- > 400-mA output current for $V_I \geq 2.0$ V, $V_O = 3.3$ V (typical 1.45-A peak switching current limit)
 - Stackable: parallel multiple devices for higher output current
- > 90% Efficiency at 10- μ A load current
 - 75-nA quiescent current
 - 60-nA shutdown current
- Single-mode operation
 - Eliminates mode transitions between buck, buck-boost and boost operation
 - Low output ripple
 - Excellent transient performance
- Safety and robust operation features
 - Integrated soft start
 - Programmable input current limit with eight settings (1 mA to 100 mA and unlimited)
 - Output short-circuit and overtemperature protection
- Tiny solution size of 21-mm²
 - Small 2.2- μ H inductor, single 22- μ F output capacitor
 - 10-Pin, 2.5-mm $\times$ 2.5-mm, 0.5-mm pitch WSON package

2 Applications

- [Smart meters and sensor nodes](#)
- [Electronic smart locks](#)
- [Medical sensor patches](#) and [patient monitors](#)
- [Wearable electronics](#)
- [Asset tracking](#)
- [Industrial IoT \(smart sensors\) / NB-IoT](#)

3 Description

The TPS63900 device is a high-efficiency synchronous buck-boost converter with an extremely low quiescent current (75 nA typical). The device has 32 user-programmable output voltage settings from 1.8 V to 5 V.

A dynamic voltage-scaling feature lets applications switch between two output voltages during operation; for example, to save power by using a lower system supply voltage during standby operation.

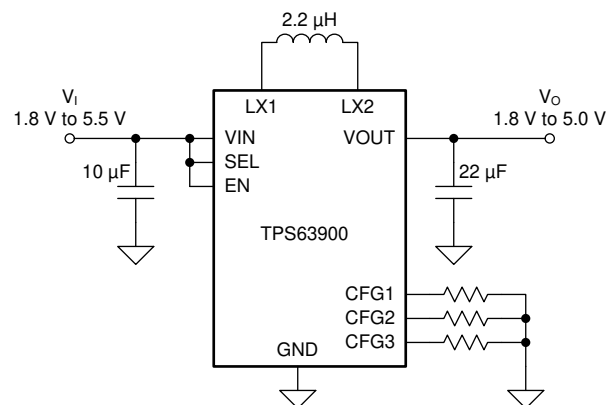
With its wide supply voltage range and programmable input current limit (1 mA to 100 mA and unlimited), the device is ideal for use with a wide range of primary like 3S Alkaline, 1S Li-MnO₂ or 1S Li-SOCl₂, and secondary battery types.

The high-output current capability supports commonly-used RF standards like sub-1-GHz, BLE, LoRa, wM-Bus, and NB-IoT.

Device Information

PART NUMBER ⁽¹⁾	PACKAGE	BODY SIZE (NOM)
TPS63900	WSON (10)	2.5 mm $\times$ 2.5 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Simplified Schematic

